



Dr. D. Y. Patil Educational Federation's

DR. D. Y. PATIL COLLEGE OF ENGINEERING AND INNOVATION



Approved by AICTE, Recognized by Govt. of Maharashtra,
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Accredited by NAAC with “A” Grade

Department of Computer Engineering

Topic: quantum vs classical microprocessor

Name: 1.neha borkar 2. mayuri mundkar 3. janvhi patil , 4. vedika karlekar

Class: Second year

Roll no :12234, 12242,12255, 12210

Div: B

Subject: Computer Architecture and Organization



Dr. D. Y. PATIL
COLLEGE OF ENGINEERING AND
INNOVATION
Varale, Talegaon, Pune Campus

- **Working of a Classical Microprocessor**

- A classical microprocessor works using **binary bits (0 and 1)** and follows the **fetch–decode–execute cycle**:

1.Fetch: It fetches an instruction from memory.

2.Decode: The Control Unit decodes the instruction.

3.Execute: The ALU performs the required operation.

4.Store: The result is stored in a register or memory.

5. This cycle repeats continuously to execute the program.

Applications of Classical Microprocessor

- **Computers and laptops** – processing data and running software.
- **Embedded systems** – used in appliances, machines, and controllers.
- **Automobiles** – engine control, sensors, and safety systems.
- **Industrial automation** – controlling machines and production systems.
- **Medical equipment** – monitoring and controlling medical devices.
- **Communication systems** – used in routers, modems, and other devices.
- **Consumer electronics** – TVs, printers, cameras, and smart devices.
- **Robotics** – controlling motors, sensors, and robotic operations.

Advantages of Classical Microprocessor

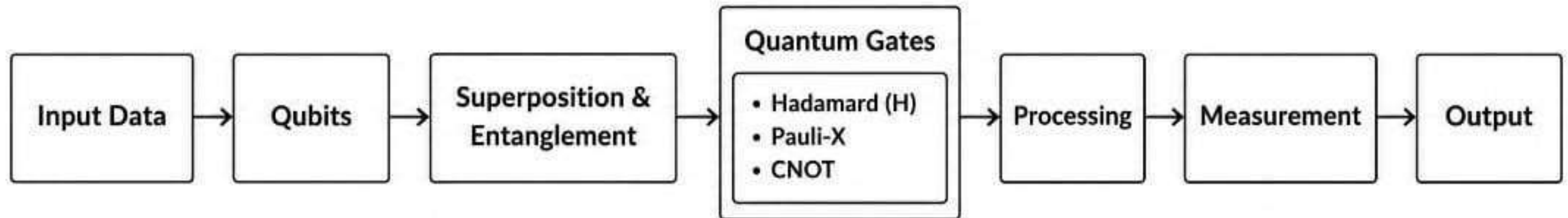
- Simple to understand and use
- Low cost
- Reliable and stable
- Easy to program
- Widely available and supported

Disadvantages of Classical Microprocessor

- Limited processing speed
- Consumes more power for complex tasks
 - Cannot perform quantum-level computations
- Performance decreases for very large problems
- Limited parallel processing capability



• Working of Quantum Microprocessor



- **Applications of Quantum Microprocessor**

1. Artificial Intelligence and Machine Learning
2. Cryptography and Cybersecurity
3. Drug and medical research
4. Weather and climate simulation
5. Optimization of complex problems

Advantages of Quantum Microprocessor

1. Very high processing power
2. Solves some complex problems faster
3. Can process multiple possibilities
4. Useful for advanced scientific research
5. Can improve optimization and simulation

Disadvantages of Quantum Microprocessor

1. Very expensive to build
2. Difficult to develop and maintain
3. Requires special operating conditions
4. Qubits are highly sensitive to errors
5. Limited practical applications currently



Aspect	Classical Microprocessor	Quantum Microprocessor
Basic Unit	Bit — holds either 0 or 1	Qubit — holds 0, 1, or both at once
State Principle	Definite binary states	Superposition & entanglement
Processing Style	Sequential / limited parallelism	Massive inherent parallelism
Speed on Hard Problems	Slows sharply as complexity grows	Exponential speed-up for specific problems
Error Rate	Very low, highly stable	High — sensitive to noise & decoherence
Operating Conditions	Room temperature	Near absolute zero (extreme cooling)
Maturity	Fully commercial, mainstream	Experimental, early-stage research

Thank You



Dr. D. Y. PATIL
COLLEGE OF ENGINEERING AND
INNOVATION
Varale, Talegaon, Pune Campus

Department of Engineering Science & Humanities